Vol 15 Chapter 1 — AC(0): Bounded-Depth Counting

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 1 — AC(0): Bounded-Depth Counting

Level 1 — one sentence

The BST team's primary methodological discipline is AC(0)-thinking — reducing every problem to bounded-depth (constant-depth) counting before invoking sophisticated mathematical machinery — operationalized via Casey's `/ac0` slash command and grounded in the substrate's AC(0)-bounded computational nature (Vol 14 Ch 9-10).

Level 2 — graduate-physicist precision

1.1 AC(0) complexity class

AC(0) = constant-depth polynomial-size circuit families with unbounded-fan-in AND, OR, NOT gates.

Negative results: - Hastad 1986: PARITY $\notin$ AC(0) - Razborov-Smolensky 1987: MAJORITY $\notin$ AC(0) - Furst-Saxe-Sipser 1984: original AC(0) parity bound

Positive results: - Addition $\in$ AC(0) - Multiplication $\in$ AC(0) (poly-size) - Most polynomial-time decidable trivia $\in$ AC(0)

1.2 BST AC(0) bounds (proved)

T421 (Depth Ceiling): substrate depth ≤ 1 under Casey strict-protocol.

T316: substrate depth $\leq$ rank $= 2$.

T317-T319 (Observer hierarchy): CI coupling depth $\leq$ rank.

Conclusion: BST substrate IS AC(0)-bounded.

1.3 The discipline in practice

First question Casey asks of any new claim: "What's the AC(0) proof?"

If you can't reduce to AC(0)-counting, you don't yet understand the claim.

`/ac0` slash command: structured AC(0) analysis — what counts, at what depth, with what circuit complexity.

Sophistication is allowed, but only AFTER AC(0) reduction is shown.

1.4 Why this works

Substrate is AC(0)-bounded (Vol 14 Ch 9-10). All physical observables reduce to bounded-depth substrate-K-type counting.

Pedagogical: any claim that resists AC(0) reduction is probably not understood. Either simplify or hold.

Anti-sophistication-bias: avoids the trap of impressive notation hiding shallow reasoning (feedback_sophistication_bias.md memory).

1.5 K-audit anchors

- T421: depth ≤ 1
- T316: depth $\leq$ rank $= 2$
- /ac0 skill (Casey standing tool)
- Vol 14 Ch 9-10: substrate AC(0)

Level 3 — 5th-grader accessibility

AC(0) = constant-depth circuits; can compute easy stuff (addition), can't compute parity. BST proved substrate is AC(0) (T421, T316). Methodological rule: "What's the AC(0) proof?" — reduce to bounded-depth counting first, sophistication second. Avoids sophistication bias (impressive notation hiding shallow reasoning).

What comes next

Chapter 2 develops the AC graph as research instrument.

Where to look this up

- Hastad 1986; Razborov-Smolensky 1987
- BST: T421, T316; `/ac0` skill